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Abstract

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Methods

Participants

In the American English validation study, the participants were 204 parents of children aged 15 to 36 months ($M = 26.21$, $Mdn = 26$, SD = 6.37). There were 103 girls in the sample (50.49%). The sample was ethnically diverse, including White, Black, Latino/Hispanic, Asian, Native American, and other races/ethnicities. None of the children were reported to hear other languages apart from English. The average number of years of education attained by the primary caregiver was 15.82 (SD = 2.27).

In the Polish validation study, the participants were 113 parents of children aged 18 to 34 months ($M = 24.73$, $Mdn = 24$, SD = 4.47). There were 49 girls in the sample (43.36%). The sample was White (reflecting the limited ethnic diversity within the Polish population). However, there were 28 children multilingual with Polish (as their home language) and some other (societal) language, living outside of Poland. Parents were asked about their education level and 65 reported having higher education or PhD, 1 reported having unfinished higher education, 11 reported having secondary education and 36 did not report their education level.

Notably, the two validation samples differed slightly (but significantly) in the mean age of children, $\Delta M = -1.49$, 95% CI $[-2.69, -0.28]$, $t(297.82) = -2.42$, $p = .016$, which may affect the outcomes of direct comparisons between the samples, where age could be a contributing factor. We present one such comparison–standard error of CAT ability–where this difference is explicitly discussed. *NOTE*: You can also view/comment on this on GitHub Issue #7 linked here.

Materials

The data was gathered with CAT-CDIs in American English and Polish. The American English CAT-CDI was created from the items that were common to three CDIs: Words and Gestures, Words and Sentences and CDI-III. With these items, two CAT-CDIs were created, one for word production and another for word comprehension (see Kachergis et al. 2022 for more details). The Polish CAT-CDIs were created following a largely similar procedure as the ones in American English, but there was a separate CDI-CAT developed for each CDI version (CDI: Words and Gestures and CDI: Words and Sentences) and for each CDI:WG subscale (word production, word comprehension, gesture use) (Krajewski et al. in preparation). The American English CAT-CDI word production included 679 items and the Polish CAT-CDI version of WS: Words and Sentences included 666 items. There are 395 words common to both CDI-CAT language versions.

Procedure

Though in both American and Polish validation studies the parents were asked to fill in both the full CDI and the CAT-CDI, the procedure differed between the two studies. In the validation of American English CDI-CAT, parents were given both CDIs in one sitting, but the order of the CDI versions was counterbalanced. In the Polish validation study, the order of the CDIs was fixed, with the full CDI always first, and then, after 1–30 days (M = 8.31, Mdn = 8, SD = 4.90) parents were asked to fill in the CAT-CDI. Both studies were done fully online and obtained the approval of adequate ethics committees.

Results

Psychometric properties of the two CAT-CDIs

Our first aim was to examine whether CAT-CDIs in American English and Polish demonstrate comparable psychometric properties. To that end, we revisit the psychometric properties reported for the American English CAT-CDI (word production) in Kachergis et al. (2022) and compare those to the data from Polish CAT-CDI:Words and Sentences.

We found similarly strong correlations in the two languages between the abilities estimated from CDI-CAT and full CDI scores (English and Polish: $r = .86$). The correlations were also strong within individual age groups (see Tables 1 and 2). We also obtained IRT ability estimates from the full CDI using the same 2-PL model. Because these estimates draw on responses to all available items in the full CDI, they incorporate more information about each child's performance, reducing random error and increasing precision. The 2-PL model further adjusts for item difficulty and discrimination, making these ability scores robust and comparable across individuals. Such ability from the full test version is often considered a baseline for comparisons with ability estimated from a short (adaptive) version. We correlated the abilities estimated from the CDI-CAT and abilities estimated from full CDI and again found strong correlations in both languages, English and Polish: $r = .92$. Finally, we also correlated the abilities estimated from the full CDI and the full CDI scores, i.e. two measures obtained from the same data. This way we examine consistency between model-based abilities and traditional scores. Again, we found strong correlations in both languages (English: $r = .95$, Polish: $r = 0.94$), confirming that both measures reflect the same underlying ability.

Table 1

English: Correlations between ability estimated by CAT-CDI and ability estimated from full CDI by children's age.

	[15,18)	[18,21)	[21,24)	[24,27)	[27,30)	[30,33)	[33,36]
r ability CAT vs full CDI	0.95	0.85	0.82	0.83	0.59	0.84	0.86
N	26	22	26	30	28	24	48

Table 2

Polish: Correlations between ability estimated by CAT-CDI and ability estimated from full CDI by children's age.

	[15,18)	[18,21)	[21,24)	[24,27)	[27,30)	[30,33)	[33,36]
r ability CAT vs full CDI	NA	0.8	0.94	0.91	0.89	0.95	NA
N	NA	29	22	16	23	22	1

Next, we looked at the mean squared error (MSE) which quantifies the difference between the abilities as estimated by CAT-CDI and the full CDI (for each participant, we subtract the CAT-CDI estimate from the full estimate and square this difference to eliminate negative values). The mean squared error in English was 0.57 (*Mdn* = 0.35, *SD* = 0.70), and in Polish it was 0.19 (*Mdn* = 0.08, *SD* = 0.45). The MSE was relatively high in English (compared to Polish), indicating a relatively larger discrepancy between CAT-CDI ability estimates and full CDI ability estimates in English. This occurred despite strong correlations between the two measures. This difference between the full and CAT-CDI estimates may be explained by at least two reasons. First, a number of participants with extreme discrepancies between the CAT-CDI and full CDI estimates may be inflating the MSE. Second, there might be a systematic bias, with one estimate consistently higher than the other.

To explore the first of these possibilities, we identified children for whom the estimates from the CAT-CDI and full CDI showed extreme discrepancies, i.e. the difference between the two estimates was removed by 1.5 SD from the mean. There were 15 such cases (7.35%) in the English dataset and 4 cases (1.96%) in the Polish dataset. We then re-calculated the mean squared error without these cases, which yielded an MSE of 0.44 (Mdn = 0.29, SD = 0.44) in English and 0.12 (Mdn = 0.07, SD = 0.14) in Polish. The updated MSE values were only slightly lower than the original ones, suggesting that extreme discrepancies had only a modest effect on the overall error and cannot fully account for the differences between ability estimates from the full CDI and the CAT-CDI in the two languages.

Next, we examined a possible bias in our two ability estimates. We found that the mean CAT ability estimates were higher than the full CDI ability estimates. That was true both in English, $M_D = 0.55$, 95% CI $[0.48, 0.62]$, $t(203) = 15.11$, $p < .001$ (large effect size: Cohen's $d = 1.06$, 95% CI $[0.89, 1.23]$), and in Polish, $M_D = 0.26$, 95% CI $[0.19, 0.32]$, $t(112) = 7.71$, $p < .001$ (medium effect size: Cohen's $d = 0.73$, 95% CI $[0.52, 0.93]$).

The question that remains is whether this consistent difference between the two estimates of abilities comes from CAT over-estimating child's true lexical ability or from the full CDI underestimating child's ability. For one, it may be that in CAT-CDI parents answer "yes, my child produces this word" to more items that would be expected (Kachergis et al. 2022), and as a result, CAT-CDI may overestimate the child's true ability. On the other hand, it could be that as the full CDI is very long, parents may omit marking some items that their child produces (see e.g., Łuniewska et al., 2025 for parental inconsistencies in reporting vocabulary of older children). This could result in the full CDI underestimating the child's true ability. It is possible that either or both are true and the

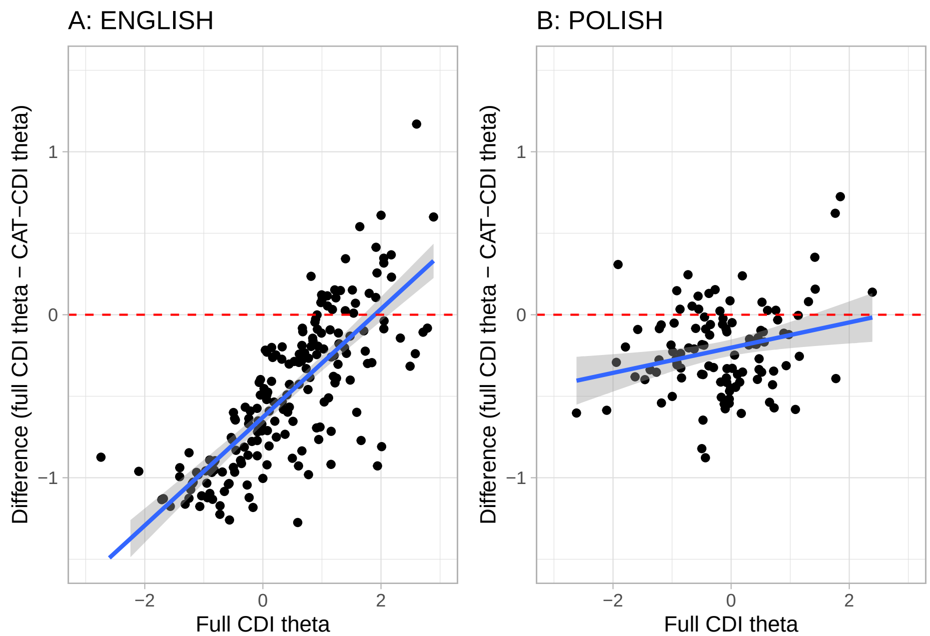

Figure 1. The relation between ability (estimated from full CDI) and the direction and size of discrepancy between the estimates from full CDI and CAT-CDI: (A) English, (B) Polish. The red dashed line at 0 indicates null difference between the two estimates. Negative values indicate that CAT estimate was higher than the full estimate. Postive values indicate that CAT estimate was lower than the full estimate.

question is difficult to answer without direct assessment of child's lexical ability. What the data tentatively suggest is that the difference between the full and CAT-CDI estimates depends on the child's ability level (see Figure 1), especially in English. For low abilities, the English CAT-CDI estimates tend to be higher than the full scores (i.e. the difference between the full estimates and CAT estimates is negative). For higher ability, the estimates from the two CDIs seem most aligned with the difference being close to zero for many of the participants. In Polish, though many negative differences are present (i.e. CAT-CDI estimate > full CDI estimate), the CAT-CDI estimates generally align more closely with the full CDI estimates across the whole range of ability.

KM's comment: These large difference between estimates visible at lowest end of ability (especially for English) may come from the fact that the difficulty of the English CDI items was calibrated on children outside the WS range (including younger kids). So if a child didn't know these "difficult" (from CAT perspective) items, the CAT estimate of their ability went higher because failing hard items implies a higher theta than failing easy ones. But the same words weren't considered "that difficult" by the full CDI, so not knowing them in full CDI indicated a relatively lower ability?

We also looked at the precision of the full and CAT-CDI estimates and how that precision (measured by standard error of the theta) relates to the ability levels. In both languages, the characteristic U-shaped curves indicate that measurement precision is highest around the middle of the ability range and declines at the extremes (very low or very high ability) (see Figure 2).

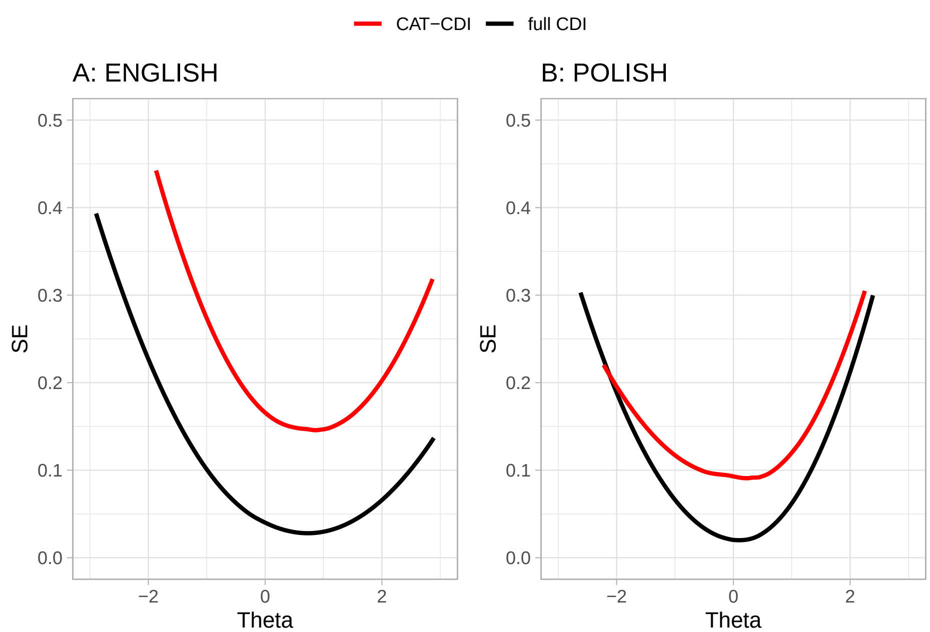

Figure 2. The relation between ability estimaes from full CDI (black) and CAT-CDI (red) and the standard error of that estimate: (A) English, (B) Polish.

In both CAT-CDIs, we have been able to reliably estimate lexical ability for majority of children. Taking as the cut-off Makransky's suggestion to consider an SE below 0.2 as acceptable, such precision of the CAT-CDI estimate was was obtained for 92.6% of the American participants, and 96.5% of the Polish participants. The remaining participants were at the extremes of the ability scale–children with either relatively low or relatively high lexical skills (see Figure 2). The ability estimates from the Polish CAT-CDI generally fell well below an SE of 0.2, except for the highest-ability observations, where precision was worse, but still similar to the precision of the full CDI. Because the Polish CAT-CDI was designed to stop once an SE of 0.1 was reached, the precision of Polish estimates never went below this threshold. Overall, the Polish CAT-CDI appears to provide precision comparable to the full CDI, particularly for children with relatively low abilities. This suggests the CAT-CDI is selecting items that are more informative at those lower ability levels.

Taken together, the results reported so far show that for the English CAT-CDI, the CAT-CDI estimates correlate strongly with both full CDI scores and estimates obtained from the full CDI. The CAT-CDI estimates are consistently higher than the full CDI and show slightly worse precision than the full CDI, however, for the large majority of children, the obtained CAT-CDI estimates were below the precision cut-off point of 0.2. In Polish, the CAT-CDI estimates also correlate strongly with both full CDI scores and estimates obtained from the full CDI. The CAT-CDI estimates are consistently higher than the full CDI, but this discrepancy is smaller than in English . The Polish CAT-CDI also yielded ability estimates with more similar precision as the full CDI, and importantly, for children with lower abilities, CAT-CDI was able to provide an estimate with the same precision as the full CDI. This might potentially suggest that for lower abilities, CAT does not over-estimate children's true abilities, however, more research needs to be done.

Item properties in the two CAT-CDIs

Our second aim was to analyze similarities and differences in IRT item properties in CAT in English and Polish.

There are 679 items in the English CAT-CDI and 666 items in the Polish CAT-CDI. In both languages, the items' difficulty and discrimination parameters were calculated using IRT 2 parameter model (these included separate samples, see Kachergis et al. 2022 and Krajewski et al., in preparation). An item's difficulty indicates the ability level at which there is a 50% probability that a participant will respond correctly (here: that a child will know a word). Difficulty is expressed on the same scale as theta (measured ability): negative values correspond to harder items, values near zero to items of moderate difficulty, and positive values to easier items (note that difficulty is inversely related to theta and so negative theta means low ability while positive theta means high ability). An item's discrimination shows how well it can distinguish between individuals whose abilities are close. In this paper, we focus mainly on item difficulty, as it is directly tied to the ability being measured, whereas discrimination reflects more the quality of the item as a measurement tool.

English items were in general more difficult than the Polish items, $\Delta M = -1.87$, 95% CI $[-2.06, -1.69]$, $t(1227.61) = -20.09$, $p < .001$ (English: min = -7.16, max = 4.45, M = -2.19, Mdn = -2.21, SD = 1.98; Polish: min = -4.34, max = 4.41, M = -0.32, Mdn = -0.43, SD = 1.41). This was true also for a subset of 390 items common to both languages - English items still proved to be more difficult than the Polish ones: $M_D = -1.68$, 95% CI $[-1.82, -1.54]$, $t(389) = -23.11$, $p < .001$. This is a result of the characteristics of the samples used to estimate the item parameters. In English, item difficulty was calculated based on a broader sample of children aged 12–36 months (spanning the CDI:WG, CDI:WS, and CDI-III), whereas the Polish data came from a sample of narrower age range of 18–36 months, corresponding to the CDI:WS. As a result, item difficulty in English was estimated using a relatively younger sample, for whom certain items may have been more challenging—thus appearing more difficult—compared to the older Polish sample. Still the item difficulty for common items in the two languages was positively and moderately correlated: $r = .65$, 95% CI $[.58, .70]$, $t(388) = 16.68$, $p < .001$.

We also examined whether CDI semantic categories show comparable difficulty values across the two languages. To this end, we conducted a series of Spearman's rank correlations on item difficulty within each semantic category (we considered only items common to both the Polish and English CDIs) (see Table 3). The rank correlations coefficients varied by category, but for half of the categories the correlations were moderate to strong (0.52 to 0.91).

Table 3

Cross-linguistic correlations of item difficulty within each CDI semantic category (Spearman's rank correlations).

CDI category	rho	n
quantifiers	0.11	10
clothing	0.22	19
connecting_words	0.23	8
locations	0.31	6
pronouns_demonstrative	0.32	4
places	0.32	7
vehicles	0.45	9
furniture_rooms	0.47	17
action_words	0.50	75
body_parts	0.52	17
descriptive_words	0.55	23
prepositions	0.55	10
games_routines	0.57	12
time_words	0.57	8
household	0.57	30
outside	0.58	20
sounds	0.60	6
food_drink	0.61	39
people	0.67	16
animals	0.68	29
toys	0.74	15
question_words	0.91	10

Last, we investigated whether particular lexical categories show related difficulty values across the two languages. We considered the following lexical classes: verbs, nouns, adjectives, function words and other, i.e. mostly sounds and words relating to routines (e.g. “hello”, “yes”, “thank you”) and time (e.g. “tommorow”, “day”). Again, we performed a series of Spearman’s rank correlations (on the difficulty of items common to CDI-CATs in Polish and English) for each lexical class (see Table 4. The weakest correlation was for function words (0.40), then verbs (0.47), adjectives (0.53), nouns (0.6), other (0.75). Notably, function words include many of the CDI semantic categories which showed the lowest correlations in Table 3, i.e., quantifiers, connecting words, locations, and pronouns. Interestingly, many English function words map onto multiple Polish equivalents - for example, “which” corresponds to “jaki” (when asking about type, kind, or quality) and “który” (when selecting from a limited set of options). These cross-linguistic differences may partly explain the weaker correlation for function words. But also, because function words are structurally tied to grammar (which is highly language-specific) they naturally vary in frequency in child-directed speech and in their communicative salience, which can affect how well children know them across languages. By contrast, the relatively high correlations for nouns reflect their consistent easiness/difficulty across languages: words like “mum”, “eye”, and “ball” are acquired early, aligning with findings that nouns are strongly and consistently over-represented in early vocabularies across many languages (e.g., Frank et al., 2021).

Table 4

lexical_class_en	rho	n
function_words	0.40	50
verbs	0.49	71
adjectives	0.53	25
nouns	0.59	194
other	0.75	50

Item selection in the two CAT-CDIs

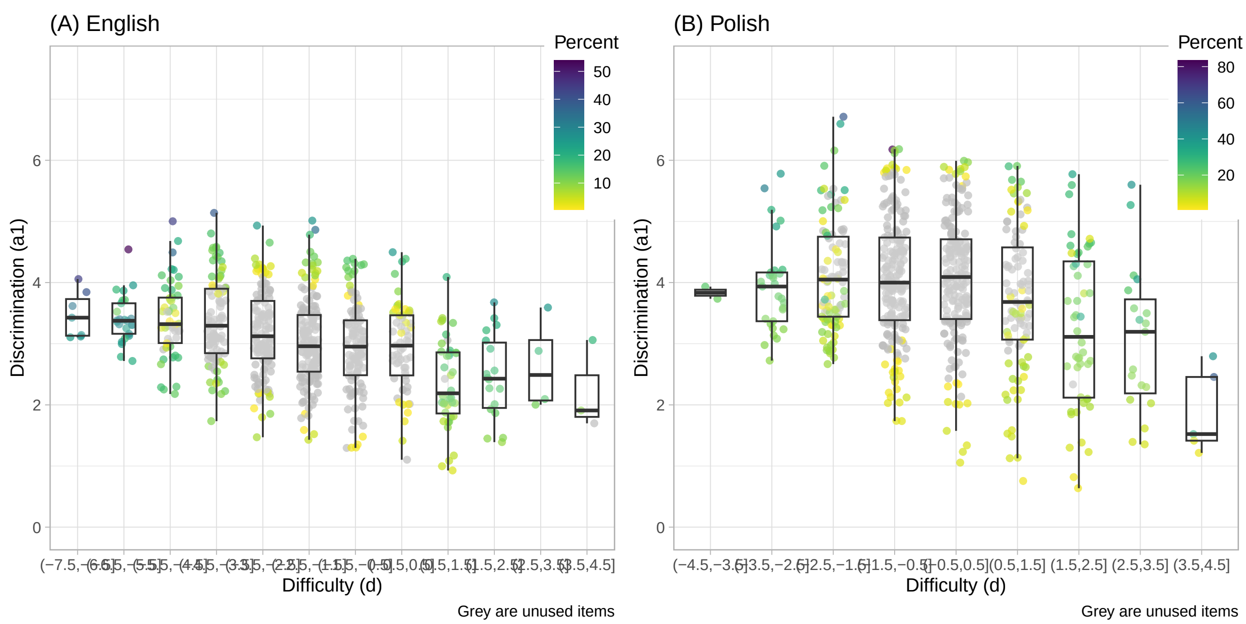

Figure 3. How often a given item was used in CAT-CDI administrations in the validation study in (A) English and (B) Polish. The items (points) are colored by the percentage of their appearance in the CAT-CDI administrations. Items colored in grey are items never used in any of the CAT-CDI administrations.

It is to be expected that a CDI-CAT will not need to administer all the items in the item bank. In fact, in the validation study the English CAT-CDI used 251 items (36.91%) and similarly, the Polish CAT-CDI used 258 items (38.74%). By design, a CDI-CAT selects items that are most informative for each participant. This means it draws from a subset of available items–typically those matched to the participant's current ability estimate in terms of difficulty, and with high discrimination, meaning they effectively distinguish between individuals with abilities close to that estimate. Figure 3 plots the items by their difficulty and discrimination parameters and colors the items (points) by how often they were used in CAT-CDI administrations in English and Polish. A few findings are of note here. First, both CAT-CDIs used items of very low discrimination (i.e, items that do no discriminate well between ability levels). Often that was because for a given difficulty level there were not enough items and CAT-CDI had to use all the items available (this is particularly true of the items on the two ends of difficulty). However, more surprisingly, it is also true of items of medium difficulty, where items of higher discrimination were available (but were not chosen by the CAT algorithm). *NOTE*: See Issue 2 here.

Finally, some items keep being shown in many CAT administrations. In English, 3 items appear in more than half of administrations - *"long"* in 64% of administrations, *"make"* in 54% of administrations, and *"last"* in 53% of administrations. They are also high discrimination items in their difficulty bins. "Long" is additionally one of the starting items, i.e. a first item shown to the parent, chosen so that there is high probability the parent can respond positively. In Polish, 3 items appear in more than half of administrations - *"szukać"* (to look for) appearing in 83% of administrations, *"znaleźć"* (to find) appearing in 61% of administrations, and *"babcia"* (grandmother) appearing in 60% of administrations. All three items show very high discrimination. "Szukać" (to look for) and "znaleźć" (to find) are also in top 5 discrimination items in general. "Babcia" (grandmother) is one of the starting items.

To check whether the CDI items common in both languages may show similar frequencies in CAT-CDI administrations, we ran Spearman's rank correlation and found that the items frequencies in administrations were weakly correlated, $r_s = .37$, $S = 6,420,909.29$, $p < .001$. However, the two CAT-CDIs differed in their potential length, as the English CAT-CDI was set to administer 25–50 items and the Polish CAT-CDI could administer up to 75 items (no minimum was set, but in practice the minimum items administered were 13). As an item is more likely to appear in longer tests (simply due to test length), then frequency of the Polish items could be inflated. *NOTE*:Could it be that, as an item is more likely to appear in longer tests (simply due to test length), frequency of the Polish items could be inflated? Could it influence the item's rank? I could probably recalculate item frequency not per total number of administrations, but per total number of items administered? SEE ISSUE #3

CAT-CDIs' usefullness in research and practice

One of the key metrics for CAT-CDI's usefulness is whether it can shorten the CDI administration compared to the full CDI, but still obtain a reliable estimate of child's lexical ability. We have found that CAT-CDIs were significantly shorter to administer: the English CAT-CDI median time was 342s (~5.7 minutes) (median full CDI time was) and the Polish CAT-CDI median time was 117s (~1.95 minutes) (median full CDI time was 1240s (~20.67 minutes)). Even though the CAT-CDI in Polish seemed shorter (compared to English CAT-CDI), the number of items administered in the two language versions of CAT-CDI was comparable: English $M = 31.42$, $Mdn = 25$ (25–50 items), Polish $M = 34.71$, $Mdn = 23$ (13–75 items).

The two CAT-CDIs differed in their settings as to when to finish the administration. Specifically, the Polish CAT-CDI was set so that the SEM as low as 0.1 be reached (with as few items as possible), and if the SEM could not be lowered to or below 0.1, the CAT-CDI stopped after administering a maximum of 75 items. The English CAT-CDI on the other hand was set to administer a maximum of 50 items, but the administration could be stopped earlier if the SEM of 0.15 or lower was reached. In other words, the Polish

CAT-CDI prioritized as low SEM as possible, while the English CAT-CDI prioritized shorter test length. This difference resulted in slightly lower mean SEM in Polish CAT-CDI, compared to English CAT-CDI $\Delta M = 0.06$, 95% CI $[0.05, 0.07]$, $t(257.25) = 11.36$, $p < .001$ but comparable mean number of items administered, $\Delta M = -3.29$, 95% CI $[-7.74, 1.16]$, $t(137.00) = -1.46$, $p = .147$.

Parental consistency in responses

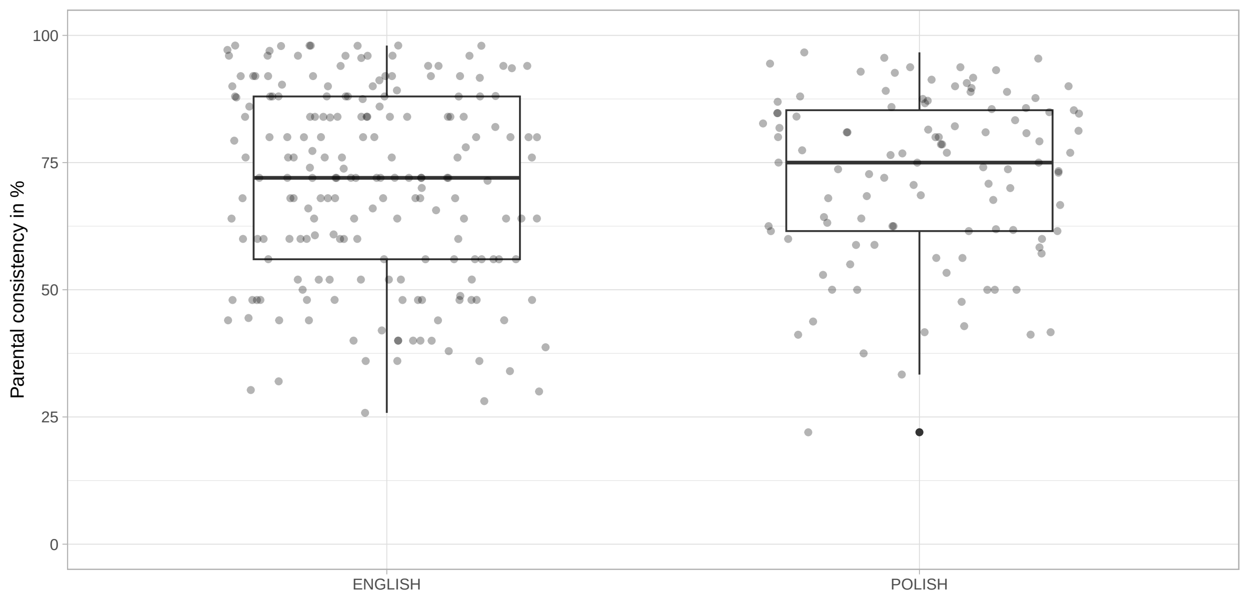

Figure 4. Percentage of parents' consistency in responding to items in the two CDI versions in the same way (in the English and in the Polish dataset).

We compared the parental consistency in their responses in the full and CAT version of CDI in two ways - using a percentage of agreement and by calculating Cohen's Kappa, which additionally factors in chance agreement. When calculating the percentage agreement, for each parent we calculated the number of items to which they responded in the same way in both CDI versions against the whole number of items asked in both CDI versions. We found though the Median consistency was relatively high: 72% for English and 75% for Polish, it varied largely across parents in both samples (see Figure 4).

We also calculated Cohen's kappa per each parent, considering their observed agreement (between responses to the same items in full and CAT CDI) and expected agreement by chance. The kappa values indicated fair (or close to fair) agreement: English $M = 0.40$, $Mdn = 0.37$, Polish $M = 0.39$, $Mdn = 0.34$. *NOTE*: There's an Issue for that here

Discussion

In points for now:

1. *Strong correlations with full CDI.*
   The CAT-CDIs were strongly correlated with the full CDI versions in both languages, which supports the overall validity of the CAT approach. These correlations held for both monolingual and multilingual participants. However, the Polish sample was small, so further research is needed—and planned—to confirm these findings.

2. *Cases of extreme divergence.*
   CAT-CDI scores diverged notably from full CDI estimates for only a small number of children in both language groups. In the Polish sample all divergent cases had low SEM, and their full CDIs were unusually short–maybe the full form underestimated their ability? In the English sample, their full CDIs were not very short, but SEM was relatively low, which is good.

3. *Item paramteres across languages.*
   Item parameters (i.e., difficulty and discrimination) showed moderate correlations across languages. These likely stem from variations in the calibration samples. Still, some semantic categories showed consistent patterns across languages, suggesting underlying structural similarities in lexical development.

4. *Item selection.*

Each CAT version used c.a. half of the full CDI item pool. Few items were overused (mostly those with highest discrimination in the item pool or starting items). But we also saw items being chosen for CAT that wouldn't be considered very informative (i.e., of low discrimination). That's something to explore further, I wrote to Phil Chalmers (author of mirt and mirtCAT), maybe he'll be able to help?

5. *Efficiency and Reliability.*

CAT-CDIs were significantly shorter than the full CDIs while maintaining high reliability for most children. The few less reliable estimates were for children at the ends of the ability range (children with either very low or very high lexical skills). Comparing Polish and English samples, the ability ranges with low SEM differed (see Figure 3) - this again might be a result of the calibration samples and item parameters? This is a good place to stress the importance of well-balanced, diverse calibration datasets. Also, while the CAT-CDI is efficient and useful for screening or initial identification ("pomiar przesiewowy" in Polish), it may be less suited for diagnostic purposes where high precision is needed across the full ability range.

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